Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	Team ID	Project Name	Maximum Marks
11 october 2026		Car Rental Request Automation	4 Marks

Technical Architecture

The solution is built entirely on the ServiceNow platform using Service Catalog, Flow Designer, Business Rules and standard tables. No external custom servers are required.

High-Level Architecture Overview

- User Interface Layer: Service Catalog form + Service Portal / My Requests views
- Process / Automation Layer: Flow Designer (primary), Business Rules, Client Scripts
- Data Layer: sc_request, sc_req_item, sc_task, sysapproval_approver + optional custom Car Inventory table
- Notification Layer: ServiceNow Notification & Email engine
- Deployment: Update Sets for configuration promotion across Dev → Test → Prod
- Infrastructure: Fully cloud-hosted ServiceNow instance (SaaS)

Table-1 : Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Service Catalog Item form for request submission Service Portal / My Requests for status tracking Backend forms for Approvers & Transport team	ServiceNow Service Catalog Service Portal UI Policies / Client Scripts
2	Application Logic – Flow	End-to-end automation: trigger on RITM creation → Manager Approval → Task creation → Notifications → Closure	ServiceNow Flow Designer Flow Triggers & Actions
3	Application Logic – Rules	Auto-populate fields, enforce mandatory data, update states after approval / task closure, set assignment groups	Business Rules Script Includes (if needed)
4	Client Scripting	Form validation, dynamic field visibility, onChange / onSubmit logic	Client Scripts UI Policies Catalog Client Scripts
5	Database / Tables	Request header & item (REQ / RITM) Fulfillment tasks (SCTASK) Approvals Optional Car Inventory custom table	sc_request, sc_req_item sc_task sysapproval_approver Custom table (u_car_inventory)
6	Notifications	Email & in-platform notifications at submission, approval, rejection, task assignment and closure	ServiceNow Notifications Email Templates Event Queue
7	SLA & Reporting	SLA definitions for approval & fulfillment stages Dashboards / reports for cycle time and open requests	SLA Definitions Performance Analytics / Reports
8	Deployment	All configuration captured and moved via Update Sets	Update Sets Retrieved Update Sets
9	Infrastructure	Fully managed ServiceNow cloud instance No local servers required	ServiceNow SaaS (Cloud) Instance: Dev / Test / Prod

Table-2 : Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source / Platform Frameworks	Leverages ServiceNow's out-of-the-box Service Catalog, Flow Designer and core tables. Custom code kept minimal.	ServiceNow Platform Flow Designer JavaScript (server & client)
2	Security Implementations	Role-based ACLs on catalog item, RITM, SCTASK and inventory table Manager can only approve their direct reports' requests Audit history on all records	ACLs Roles & Groups sys_audit
3	Scalable Architecture	3-tier logical separation (UI / Process / Data) within ServiceNow	ServiceNow multi-instance architecture Flow Designer

		Flow Designer scales with instance capacity Easy to add new car types, locations or approval levels	Update Sets
4	Availability	Inherits ServiceNow platform high availability and SLA No custom single points of failure introduced	ServiceNow Cloud HA Instance redundancy
5	Performance	Lightweight catalog form and client scripts Efficient Flow steps and GlideRecord usage Notifications queued asynchronously	Client Scripts best practices Flow Designer Event Queue

References

- ServiceNow Service Catalog & Flow Designer documentation – https://docs.servicenow.com/
- C4 Model for architecture diagrams – https://c4model.com/
- ServiceNow Update Set best practices